DIPLOMA WORK · ASTANA IT UNIVERSITY · 6B06103 BIG DATA ANALYSIS · 2027

# Data-Driven Framework for Spatial Accessibility Assessment and Optimal Siting of Social Infrastructure

Walking access from every home in Astana to every state school and preschool, built from open data and checked against official replies to information requests; and a test of where the city places its new schools.

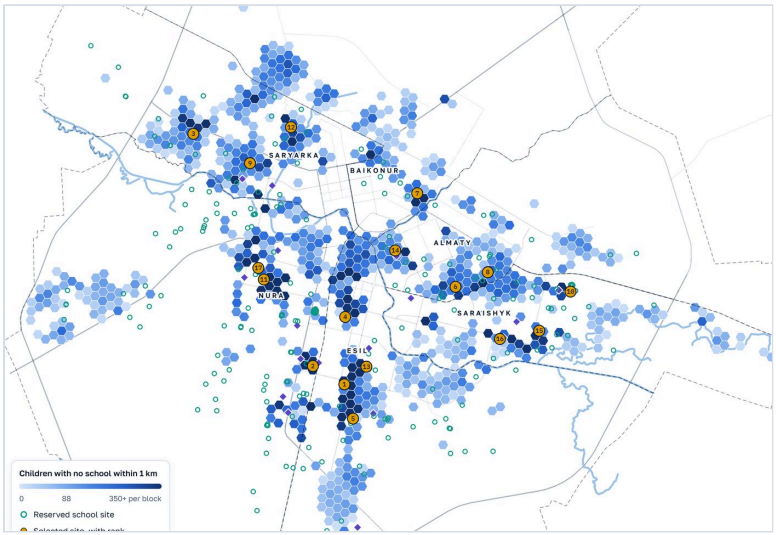
Interactive atlas

**22%**
of school-age children live within the statutory 500 m of a school on foot; 64% as the crow flies

**76%**
of the city's mean access to schools in Nura, the fastest-growing district; Esil 81%

**15%**
of the 77,870 children without a school within 1 km reached by the 17 new schools of the 2026–2028 plan

**2.4×**
as many reached by the 17 best of 238 sites already reserved in the detailed plans; 1.9–3.4× in 90% of Monte Carlo draws




Site screening in the atlas: blue blocks are children with no state school within 1 km on foot; green rings are reserved school sites, gold the 17 that reach the most children with their rank, violet diamonds the new schools of the 2026–2028 plan.

## Hypotheses

| | | |
|---|---|---|
| **H1** | Open data estimate population below the district better than WorldPop: district error 10% against 26% | **Supported** |
| **H2** | Deficits concentrate in growth districts: Nura and Esil worst for schools, but 92% of inequality lies within districts | **In part** |
| **H3** | The pattern is robust: same worst-served cells under every building-based surface (overlap 0.8); Nura or Esil worst in 199 of 200 draws | **Cells** |
| **H4** | Comfortable School sites follow free land (ROC-AUC 0.71) more than deficit (0.59); 84% of the optimum on free land | **Direction** |

## Two more findings

- Schools are crowded as well as far: 1.7 pupils per design place, 45 of 120 schools beyond two full shifts, most in Saraishyk.
- Preschools have places, 73,419 for 66,574 children, but only 34% of children live within 300 m on foot.
- By 2028 the children without a school within 1 km grow to 84–106 thousand; the plan reaches 15–17% of them.

**1 · Population**
Official district totals over 386 polling-station territories and dwellings found by LightGBM (ROC-AUC 0.982)

**2 · Walking**
Distances from 39,735 dwellings on a 152,491-node street network

**3 · Access**
Floating catchment index, each child counted once; design capacity as supply

**4 · Equity**
Gini, Theil split by district, local Moran clusters, statutory compliance

**5 · Siting**
Capacitated maximal covering on free land; 2023 backtest; ranking of reserved sites

## Recommendations for the city

1. Measure the statutory radius on foot, not as a circle.
2. Plan by blocks: the deficit lies within districts.
3. Reserve land for schools before housing surrounds it.
4. Build first the reserved sites that reach the most children without a school; the atlas ranks all 238.

## Data and reproducibility

OpenStreetMap, Overture buildings, the akim's decision on polling stations, Bureau of National Statistics age tables; official replies of the education and construction departments; INDIGO preschool register and city geoportal compiled by the project team. One command runs the 13-stage pipeline; 108 unit tests.

Dias Bolatov, Daulet Dandykulov, Arsen Baktygaliyev · Supervisor: Aidana Zhalgas

Real data only · figures as of 1 January 2026